

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.088 / 0.0166316 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-22 19:00 UTC
Location	29.5831, -95.0156
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	6 / 667	29.01 to 100; mean 71.2211	44.81 at 2026-07-22 18:55Z; 5 min before event
asos	Air temperature (temperature)	degC	6 / 667	23.2222 to 39; mean 29.9494	37 at 2026-07-22 18:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	6 / 759	0 to 360; mean 175.837	310 at 2026-07-22 18:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	6 / 783	0 to 8.74555; mean 2.9454	3.08666 at 2026-07-22 18:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	6 / 72	5915.17 to 5959.62; mean 5933.16	5935.62 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	6 / 72	17.7446 to 2017.14; mean 587.765	1612.34 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	6 / 72	29.7729 to 67.4723; mean 49.8948	55.941 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	6 / 72	1004.59 to 1014.29; mean 1009.09	1009.5 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	6 / 72	17.875 to 24.3609; mean 21.2647	21.7781 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	6 / 72	-4.5306 to 4.0994; mean 0.7762	0.121985 at 2026-07-22 18:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	6 / 72	-6.0398 to 5.78799; mean -0.288	-2.02513 at 2026-07-22 18:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	15 / 1011	-0.004 to 0.092; mean 0.0274	0.088 at 2026-07-22 19:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 178	3 to 98; mean 37.0449	73 at 2026-07-22 19:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	64 / 2493	0 to 34.1; mean 9.3529	10.2 at 2026-07-22 19:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	3 / 216	0 to 100; mean 47.1944	17 at 2026-07-22 19:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	3 / 216	43 to 96; mean 73.7407	50 at 2026-07-22 19:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	3 / 216	0 to 11.53; mean 0.2961	0 at 2026-07-22 19:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	3 / 216	1007 to 1015; mean 1010.44	1009 at 2026-07-22 19:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	3 / 216	24.14 to 40.07; mean 30.6104	38.57 at 2026-07-22 19:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	3 / 216	0 to 360; mean 189.287	26 at 2026-07-22 19:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	3 / 216	0.4 to 7.61; mean 2.6884	3.6 at 2026-07-22 19:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	59 / 4244	0 to 149.3; mean 11.4354	6.8 at 2026-07-22 19:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; 14 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; 14 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0233633 to 0.0312825; mean 0.0271	0.0312825 at 2026-07-22 18:45Z; 14 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; 14 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000163343 to 0.000396712; mean 0.0002	0.000396712 at 2026-07-22 19:14Z; 15 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 15 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	2 / 2	5.0358e-05 to 5.87309e-05; mean 0.0001	5.0358e-05 at 2026-07-22 19:14Z; 15 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 15 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; 14 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-2.41161e-06 to 9.61895e-05; mean 0	-2.41161e-06 at 2026-07-22 19:14Z; 15 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 15 min after event
tceq	Carbon monoxide (co)	ppm	3 / 195	0 to 0.5; mean 0.1436	0.1 at 2026-07-22 19:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	11 / 671	-2.6 to 44.3; mean 6.3422	5.4 at 2026-07-22 19:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 191	-1 to 1.6; mean -0.2068	1.2 at 2026-07-22 19:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-21 06Z 93.19, 12Z 73.74, 18Z 39.69, 07-22 00Z 60.95, 06Z 79.14, 12Z 59.63, 18Z 42.15, 07-23 00Z 70.41, 06Z 75.71, 12Z 74.58, 18Z 86.64, 07-24 00Z 92.69, 06Z 87.93
asos	Air temperature (temperature)	07-21 06Z 25.03, 12Z 29.02, 18Z 35.4, 07-22 00Z 30.81, 06Z 27.01, 12Z 31.64, 18Z 37.78, 07-23 00Z 31.34, 06Z 29.2, 12Z 29.13, 18Z 27.92, 07-24 00Z 25.98, 06Z 25.79
asos	Wind direction (wind_direction)	07-21 06Z 183.4, 12Z 261.5, 18Z 218.6, 07-22 00Z 201.6, 06Z 237, 12Z 289.8, 18Z 212.8, 07-23 00Z 118.9, 06Z 45, 12Z 58.03, 18Z 159.8, 07-24 00Z 151.4, 06Z 108.2
asos	Wind speed (wind_speed)	07-21 06Z 2.077, 12Z 3.508, 18Z 2.36, 07-22 00Z 2.476, 06Z 2.982, 12Z 3.928, 18Z 2.666, 07-23 00Z 1.972, 06Z 3, 12Z 4.49, 18Z 2.557, 07-24 00Z 3.219, 06Z 2.432
noaa_gfs	500-hPa geopotential height (gh_500)	07-21 12Z 5916, 18Z 5931, 07-22 00Z 5929, 06Z 5934, 12Z 5926, 18Z 5938, 07-23 00Z 5927, 06Z 5941, 12Z 5925, 18Z 5933, 07-24 00Z 5940, 06Z 5957
noaa_gfs	Planetary boundary layer height (pbl_height)	07-21 12Z 184.9, 18Z 1188, 07-22 00Z 545.7, 06Z 109.9, 12Z 219.3, 18Z 1663, 07-23 00Z 314.4, 06Z 306.1, 12Z 329.8, 18Z 1188, 07-24 00Z 528.8, 06Z 475.5
noaa_gfs	Precipitable water (precipitable_water)	07-21 12Z 31.33, 18Z 35.78, 07-22 00Z 40.72, 06Z 45.72, 12Z 52.55, 18Z 55.8, 07-23 00Z 61.66, 06Z 46.98, 12Z 51.37, 18Z 64.06, 07-24 00Z 63.55, 06Z 49.23
noaa_gfs	Surface pressure (surface_pressure)	07-21 12Z 1011, 18Z 1011, 07-22 00Z 1008, 06Z 1009, 12Z 1009, 18Z 1009, 07-23 00Z 1006, 06Z 1008, 12Z 1008, 18Z 1008, 07-24 00Z 1009, 06Z 1013
noaa_gfs	850-hPa temperature (t_850)	07-21 12Z 22.46, 18Z 22.28, 07-22 00Z 21.85, 06Z 22.4, 12Z 21.68, 18Z 21.72, 07-23 00Z 23.75, 06Z 21.61, 12Z 20.18, 18Z 19.34, 07-24 00Z 19.88, 06Z 18.04
noaa_gfs	10-m eastward wind (u_10m)	07-21 12Z 2.739, 18Z 2.308, 07-22 00Z 0.8438, 06Z 2.517, 12Z 3.085, 18Z 0.662, 07-23 00Z -0.1673, 06Z -1.059, 12Z -0.3388, 18Z -1.688, 07-24 00Z 1.223, 06Z -0.8123
noaa_gfs	10-m northward wind (v_10m)	07-21 12Z 0.2999, 18Z -0.7365, 07-22 00Z 0.5354, 06Z 1.391, 12Z -0.4914, 18Z -2.383, 07-23 00Z 1.481, 06Z -3.055, 12Z -3.632, 18Z -5.125, 07-24 00Z 4.78, 06Z 3.481
openaq	Ozone (ozone)	07-21 06Z 0.006118, 12Z 0.01152, 18Z 0.04518, 07-22 00Z 0.02902, 06Z 0.00619, 12Z 0.01574, 18Z 0.06606, 07-23 00Z 0.03856, 06Z 0.02697, 12Z 0.02675, 18Z 0.02681, 07-24 00Z 0.02054, 06Z 0.01633
openaq	PM10 (pm10)	07-21 06Z 39.3, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 25.22, 18Z 16.39, 07-24 00Z 19.61, 06Z 36.67
openaq	PM2.5 (pm25)	07-21 06Z 7.323, 12Z 6.929, 18Z 5.735, 07-22 00Z 5.895, 06Z 6.012, 12Z 6.28, 18Z 9.156, 07-23 00Z 10.76, 06Z 17.85, 12Z 17.49, 18Z 8.554, 07-24 00Z 8.028, 06Z 13.58
openweather	Cloud cover (cloud_cover)	07-21 06Z 0, 12Z 0.7059, 18Z 1.474, 07-22 00Z 4.389, 06Z 25.94, 12Z 13.06, 18Z 63.39, 07-23 00Z 80.88, 06Z 74.79, 12Z 99.22, 18Z 99, 07-24 00Z 95.06, 06Z 78.5
openweather	Relative humidity (humidity)	07-21 06Z 90.94, 12Z 80.12, 18Z 48.68, 07-22 00Z 62.89, 06Z 80.11, 12Z 70.67, 18Z 49.17, 07-23 00Z 74.12, 06Z 77.89, 12Z 78.33, 18Z 80, 07-24 00Z 93.28, 06Z 93
openweather	Precipitation rate (precipitation)	07-21 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.01, 07-23 00Z 2.265, 06Z 0, 12Z 0, 18Z 0.6411, 07-24 00Z 0.6922, 06Z 0.635
openweather	Surface pressure (pressure)	07-21 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1009, 12Z 1010, 18Z 1009, 07-24 00Z 1012, 06Z 1014
openweather	Air temperature (temperature)	07-21 06Z 25.58, 12Z 28.88, 18Z 36.56, 07-22 00Z 31.93, 06Z 27.36, 12Z 31.33, 18Z 39.05, 07-23 00Z 31.87, 06Z 29.81, 12Z 29.42, 18Z 29.4, 07-24 00Z 25.86, 06Z 25.3
openweather	Wind direction (wind_direction)	07-21 06Z 179.7, 12Z 268.3, 18Z 255.6, 07-22 00Z 224.1, 06Z 258.9, 12Z 283.7, 18Z 172.4, 07-23 00Z 221, 06Z 89.32, 12Z 63.67, 18Z 99.06, 07-24 00Z 165.2, 06Z 167
openweather	Wind speed (wind_speed)	07-21 06Z 2.419, 12Z 2.092, 18Z 1.464, 07-22 00Z 2.726, 06Z 2.276, 12Z 2.136, 18Z 2.466, 07-23 00Z 2.711, 06Z 2.738, 12Z 3.544, 18Z 4.686, 07-24 00Z 2.944, 06Z 3.245

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-21 06Z 9.591, 12Z 8.39, 18Z 7.889, 07-22 00Z 9.967, 06Z 6.817, 12Z 7.95, 18Z 11.08, 07-23 00Z 14.7, 06Z 22.34, 12Z 17.32, 18Z 9.85, 07-24 00Z 10.33, 06Z 15.13
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CO column (s5p_co_column)	07-21 18Z 0.02656, 07-22 18Z 0.03123, 07-23 18Z 0.02345
sentinel5p	CO granules available (s5p_co_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-21 18Z 0.0002243, 07-22 18Z 0.0003967, 07-23 18Z 0.0001668
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06, 07-23 18Z 5.971e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
tceq	Carbon monoxide (co)	07-21 06Z 0.1267, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1333, 18Z 0.1611, 07-24 00Z 0.22, 06Z 0.1167
tceq	Nitrogen dioxide (no2)	07-21 06Z 5.508, 12Z 5.514, 18Z 4.918, 07-22 00Z 10.8, 06Z 6.498, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.194, 06Z 5.151, 12Z 6.013, 18Z 6.791, 07-24 00Z 7.521, 06Z 4.65
tceq	Sulfur dioxide (so2)	07-21 06Z -0.28, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.2333, 18Z -0.3278, 07-24 00Z -0.27, 06Z -0.35

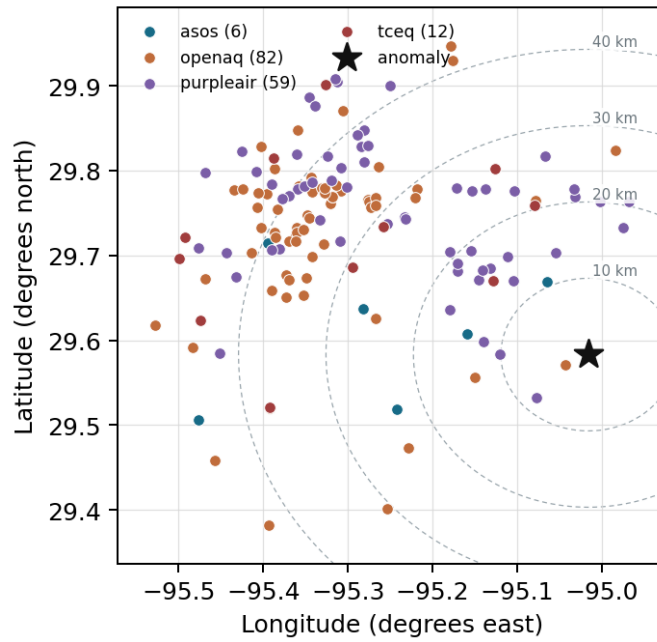
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

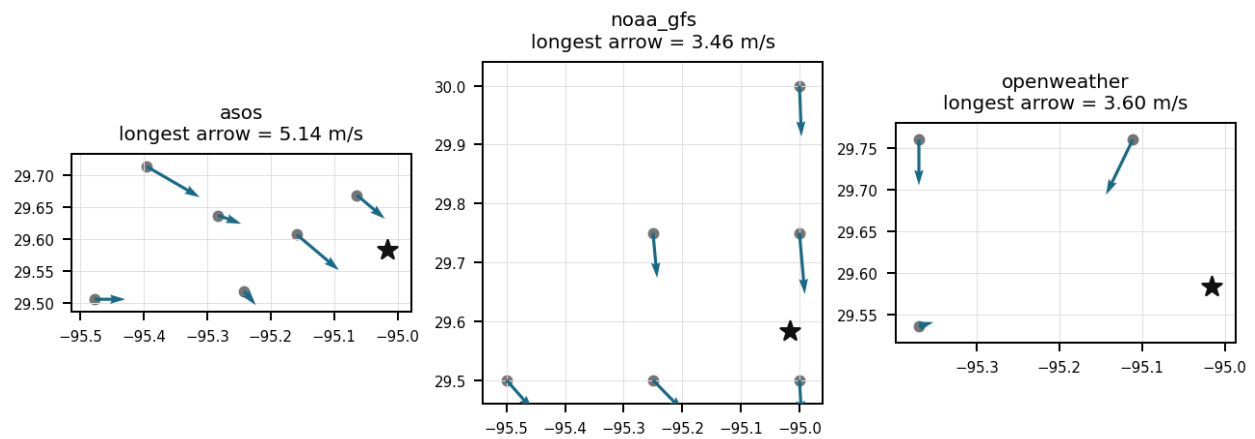
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (10.66 km) 72.29, EFD (14.095 km) 70.67, LVJ (23.006 km) 75.3, HOU (26.492 km) 71.27, MCJ (39.448 km) 67.2, AXH (45.438 km) 71.91
asos	Air temperature (temperature)	T41 (10.66 km) 29.96, EFD (14.095 km) 30.31, LVJ (23.006 km) 29.3, HOU (26.492 km) 30.06, MCJ (39.448 km) 30.59, AXH (45.438 km) 29.46
asos	Wind direction (wind_direction)	T41 (10.66 km) 190.3, EFD (14.095 km) 180.9, LVJ (23.006 km) 165.3, HOU (26.492 km) 183.1, MCJ (39.448 km) 197.6, AXH (45.438 km) 144.1
asos	Wind speed (wind_speed)	T41 (10.66 km) 3.183, EFD (14.095 km) 3.451, LVJ (23.006 km) 1.993, HOU (26.492 km) 3.58, MCJ (39.448 km) 4.13, AXH (45.438 km) 1.612
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.00 (9.363 km) 5931, gfs:29.75,-95.00 (18.62 km) 5934, gfs:29.50,-95.25 (24.486 km) 5931, gfs:29.75,-95.25 (29.28 km) 5934, gfs:30.00,-95.00 (46.382 km) 5936, gfs:29.50,-95.50 (47.763 km) 5932
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.00 (9.363 km) 538.8, gfs:29.75,-95.00 (18.62 km) 591.1, gfs:29.50,-95.25 (24.486 km) 556.4, gfs:29.75,-95.25 (29.28 km) 863.1, gfs:30.00,-95.00 (46.382 km) 434.7, gfs:29.50,-95.50 (47.763 km) 542.5
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.00 (9.363 km) 50.8, gfs:29.75,-95.00 (18.62 km) 50.18, gfs:29.50,-95.25 (24.486 km) 50.31, gfs:29.75,-95.25 (29.28 km) 49.3, gfs:30.00,-95.00 (46.382 km) 49.81, gfs:29.50,-95.50 (47.763 km) 48.96

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.00 (9.363 km) 1010, gfs:29.75,-95.00 (18.62 km) 1010, gfs:29.50,-95.25 (24.486 km) 1009, gfs:29.75,-95.25 (29.28 km) 1009, gfs:30.00,-95.00 (46.382 km) 1008, gfs:29.50,-95.50 (47.763 km) 1008
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.00 (9.363 km) 21.12, gfs:29.75,-95.00 (18.62 km) 21.3, gfs:29.50,-95.25 (24.486 km) 21.15, gfs:29.75,-95.25 (29.28 km) 21.33, gfs:30.00,-95.00 (46.382 km) 21.44, gfs:29.50,-95.50 (47.763 km) 21.25
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.00 (9.363 km) 1.151, gfs:29.75,-95.00 (18.62 km) 0.3923, gfs:29.50,-95.25 (24.486 km) 0.7273, gfs:29.75,-95.25 (29.28 km) 0.4448, gfs:30.00,-95.00 (46.382 km) 0.4023, gfs:29.50,-95.50 (47.763 km) 1.54
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.00 (9.363 km) 0.4249, gfs:29.75,-95.00 (18.62 km) -0.1517, gfs:29.50,-95.25 (24.486 km) -0.1709, gfs:29.75,-95.25 (29.28 km) -0.5792, gfs:30.00,-95.00 (46.382 km) -0.9509, gfs:29.50,-95.50 (47.763 km) -0.3001
openaq	Ozone (ozone)	5077791 (0.0 km) 0.02484, 332 (14.578 km) 0.03371, 2800 (21.045 km) 0.02357, 320 (24.779 km) 0.02815, 3214 (26.61 km) 0.02791, 339 (26.887 km) 0.02982, 2039 (28.555 km) 0.02512, 3378 (28.767 km) 0.02825, +7 more
openaq	PM10 (pm10)	271 (14.578 km) 44.44, 13380082 (28.767 km) 34.43, 15852419 (35.779 km) 30.83
openaq	PM2.5 (pm25)	14404954 (2.966 km) 9.159, 13955815 (13.303 km) 7.753, 268 (14.578 km) 15.92, 11638043 (23.947 km) 8.072, 14157975 (24.782 km) 9.895, 3379 (28.767 km) 10.63, 15539682 (28.768 km) 10.16, 8967123 (28.822 km) 11.02, +56 more
openweather	Cloud cover (cloud_cover)	grid:east (21.767 km) 44.26, grid:south (34.66 km) 50.92, grid:center (39.494 km) 46.4
openweather	Relative humidity (humidity)	grid:east (21.767 km) 77.22, grid:south (34.66 km) 73.29, grid:center (39.494 km) 70.71
openweather	Precipitation rate (precipitation)	grid:east (21.767 km) 0.2751, grid:south (34.66 km) 0.404, grid:center (39.494 km) 0.2092
openweather	Surface pressure (pressure)	grid:east (21.767 km) 1010, grid:south (34.66 km) 1010, grid:center (39.494 km) 1010
openweather	Air temperature (temperature)	grid:east (21.767 km) 30.13, grid:south (34.66 km) 30.36, grid:center (39.494 km) 31.35
openweather	Wind direction (wind_direction)	grid:east (21.767 km) 183.1, grid:south (34.66 km) 188.1, grid:center (39.494 km) 196.7
openweather	Wind speed (wind_speed)	grid:east (21.767 km) 3.235, grid:south (34.66 km) 2.457, grid:center (39.494 km) 2.373
purpleair	PM2.5 (pm25)	2386 (8.161 km) 0.5181, 3298 (10.104 km) 12.59, 247283 (12.088 km) 12.7, 268165 (12.918 km) 13.56, 222815 (13.885 km) 11.74, 222593 (15.857 km) 12.19, 128007 (15.88 km) 6.247, 161689 (15.935 km) 12.4, +51 more
tceq	Carbon monoxide (co)	482011039 (14.558 km) 0.09538, 482011035 (28.764 km) 0.1672, 482011052 (44.204 km) 0.1682
tceq	Nitrogen dioxide (no2)	482011050 (0.008 km) 4.117, 482011039 (14.558 km) 7.495, 482011015 (20.5 km) 8.597, 482010026 (26.626 km) 7.075, 482011035 (28.764 km) 7.012, 482010416 (29.317 km) 6.567, 480391004 (37.118 km) 4.888, 482011052 (44.204 km) 3.948, +3 more
tceq	Sulfur dioxide (so2)	482011039 (14.558 km) 0.3185, 482011035 (28.764 km) -0.4794, 482010051 (44.584 km) -0.4762

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

An ASOS weather station 10.66 km from the anomaly recorded a high surface temperature of 37.0 degC and relative humidity of 44.81 percent at 18:55:00 UTC on July 22, 2026.

Mark exactly one:

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Note (optional):

Claim 2 of 36

Elevated concentrations of volatile organic compound precursors, evidenced by elevated Sentinel-5P formaldehyde column density measurements, provided the necessary chemical reactivity to drive rapid tropospheric ozone formation.

Mark exactly one:

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Note (optional):

Claim 3 of 36

Hot, dry, sunny afternoon conditions were strongly favorable for rapid surface-ozone production near eastern Houston at the anomaly time, because air temperature was about 37 to 39 degrees C, relative humidity was about 45 to 50 percent, cloud cover near the anomaly was only 17 percent, and no precipitation was occurring.

Mark exactly one:

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Note (optional):

Claim 4 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

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Note (optional):

Claim 5 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:

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Note (optional):

Claim 6 of 36

Light surface winds and an expanding afternoon planetary boundary layer exceeding 1600 meters promoted local precursor accumulation and thermal mixing that enhanced afternoon ozone production.

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Note (optional):

Claim 7 of 36

Transport or recirculation of urban-industrial precursor emissions into eastern Houston was a likely contributor to the ozone spike, because winds were variable but included northwesterly flow at the nearest surface station near the event while satellite columns showed elevated carbon monoxide and strongly elevated formaldehyde with no corresponding sulfur dioxide enhancement.

Mark exactly one:

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Note (optional):

Claim 8 of 36

Chemical observations near eastern Houston on 2026-07-22 are most consistent with VOC-rich photochemical ozone production and transport because ozone reached 0.088 ppm while satellite formaldehyde and carbon monoxide were elevated, local nitrogen dioxide was only moderate, sulfur dioxide was not elevated, and particulate matter did not show a major smoke signature.

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Note (optional):

Claim 9 of 36

Satellite measurements detected a peak in atmospheric formaldehyde column density of 0.000397 mol/m² on July 22, 2026, pointing to strong organic precursor availability during the event.

Mark exactly one:

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Note (optional):

Claim 10 of 36

The PM2.5 levels are elevated at 6.8 ug/m3, which is higher than the mean of 11.4354 ug/m3.

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Note (optional):

Claim 11 of 36

The air quality monitoring data from multiple sources, including PurpleAir and TCEQ, confirmed the anomaly and provided valuable insights into its characteristics.

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Note (optional):

Claim 12 of 36

Boundary-layer structure near eastern Houston at 18:00 UTC on 2026-07-22 favored efficient entrainment and mixing of ozone and its precursors because the planetary boundary layer height was about 1612 m under relatively high 500 hPa heights.

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Note (optional):

Claim 13 of 36

High surface temperatures exceeding 37 degC combined with clear conditions on July 22, 2026, significantly accelerated photochemical reaction rates to yield elevated ground-level ozone concentrations in the Houston area.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 36

A deep planetary boundary layer height reaching approximately 1663 meters near peak sunlight facilitated the vertical mixing of photochemical precursors over the Houston region.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 36

The 6-hour mean openaq ozone concentration for 2026-07-22 18Z was 0.06606 ppm, which was substantially higher than the preceding 2026-07-22 12Z mean of 0.01574 ppm within the 50 km context window.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 16 of 36

Sentinel-5P satellite observations detected an elevated formaldehyde (HCHO) total column density of 0.000397 mol/m² over the region on July 22, 2026, at 19:14:34 UTC.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 36

Deep daytime mixing likely supported the ozone spike near eastern Houston by distributing ozone and precursors through a well-developed boundary layer, because modeled planetary boundary-layer height near the site was about 1612 m at 18Z on 2026-07-22 and 500 hPa heights were relatively high near 5936 m.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 36

Numerical weather model data indicated a deep planetary boundary layer height exceeding 1,600 meters on the afternoon of July 22, 2026, supporting intense daytime boundary layer development and photochemical reaction volume.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 20 of 36

Surface meteorology near eastern Houston at 19:00 UTC on 2026-07-22 strongly favored rapid ozone formation because air temperature was about 37 to 39 C, relative humidity was about 45 to 50 percent, cloud cover was low near the anomaly time, and no precipitation was occurring.

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Note (optional):

Claim 21 of 36

The air quality anomaly was characterized by elevated levels of PM2.5 and NO2.

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Note (optional):

Claim 22 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 36

Strong photochemical ozone production in eastern Houston was likely enhanced on 2026-07-22 by very hot afternoon temperatures near 37 to 39 degrees Celsius, relatively low humidity near 45 to 50 percent, little precipitation, and low cloud cover near the anomaly time.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 24 of 36

A deep and well-mixed afternoon boundary layer over Houston on 2026-07-22 likely promoted rapid ozone formation and downward mixing of ozone-rich air, because the planetary boundary layer height was about 1612 meters near 18Z under elevated 500 hPa heights.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 36

High ambient temperatures exceeding 37 degrees Celsius combined with low relative humidity created ideal atmospheric conditions for rapid photochemical ozone production in Houston on July 22, 2026.

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Note (optional):

Claim 26 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

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Note (optional):

Claim 27 of 36

Transport or recirculation of VOC-rich urban-industrial emissions is supported more strongly than a fresh sulfur-rich or combustion-dominated plume, because Sentinel-5P formaldehyde and carbon monoxide columns were elevated on 2026-07-22 while NO₂ near the surface was only modest at 5.4 ppb, TCEQ CO was low at 0.1 ppm, and both Sentinel-5P SO₂ and surface SO₂ remained low near the anomaly time.

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Note (optional):

Claim 28 of 36

The anomaly was likely caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 29 of 36

Satellite observations revealed a notable spike in atmospheric formaldehyde column density on July 22, 2026, providing evidence of elevated volatile organic compound concentrations that acted as ozone precursors.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 36

The openaq ozone anomaly at (29.583, -95.016) had an expected value of 0.016631578947368424 ppm and a z-score of 5.812520004185725, which is labeled severe in the provided data.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 31 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 32 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 36

The anomalous pollutant was ozone measured by openaq at 0.088 ppm at 2026-07-22T19:00:00+00:00 at coordinates (29.583, -95.016).

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 34 of 36

Near-surface atmospheric temperatures reached up to 39.0°C to 40.07°C alongside relative humidity dropping below 45% around 19:00 UTC on July 22, 2026, providing thermal conditions conducive to photochemical ozone production.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 36

The anomaly occurred on July 22nd at around 19:00 UTC, with the nearest sensor located approximately 37.201 km away.

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Note (optional):

Claim 36 of 36

An OpenAQ sensor recorded a severe ozone anomaly of 0.088 ppm at coordinates (29.583, -95.016) on July 22, 2026, at 19:00:00 UTC, compared to an expected baseline of approximately 0.0166 ppm (z-score 5.81).

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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
